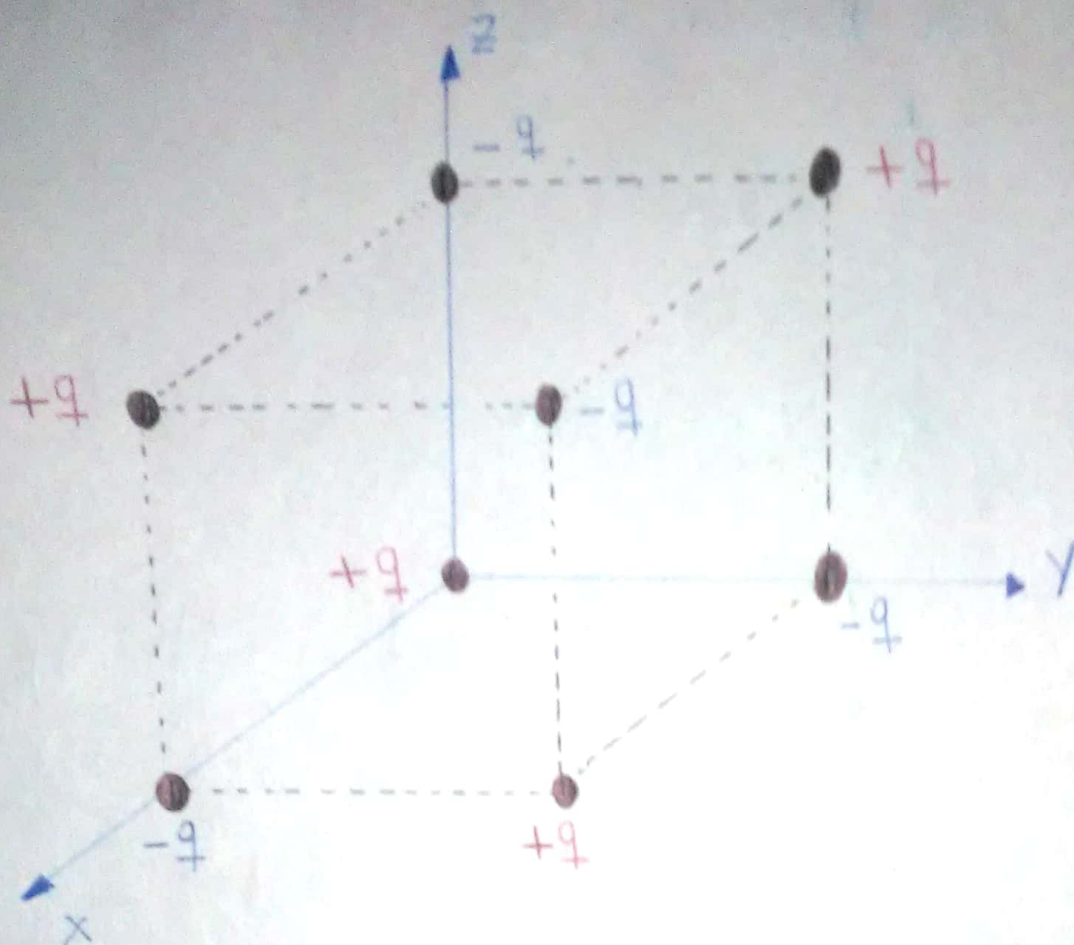


1) Sea la distribución de cargas.

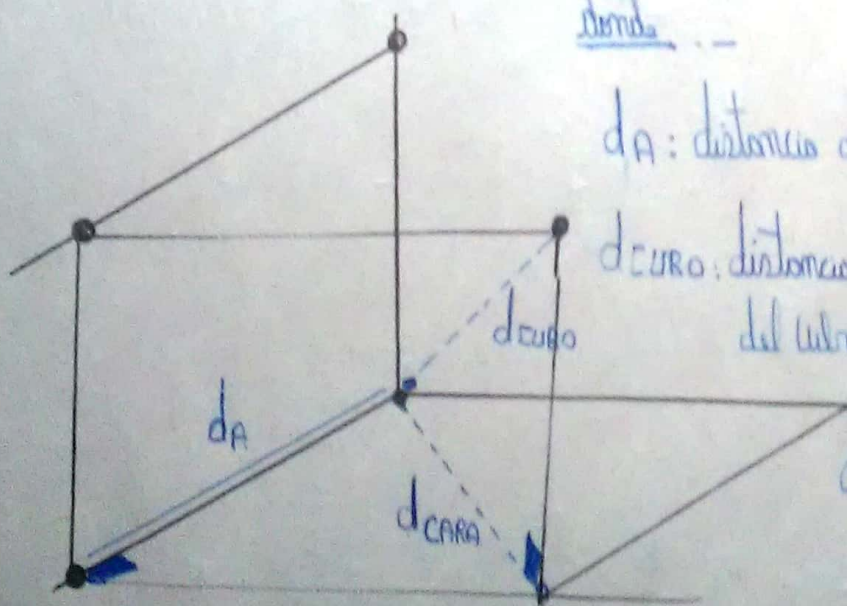


Consideremos las siguientes medidas.

donde

d_A : distancia arista $\Rightarrow \underline{d_A = a}$

d_{CUBO} : distancia diagonal del cubo $\Rightarrow \underline{d_{CUBO} = \sqrt{3}a}$



d_{CARA} : distancia diagonal de una cara $\Rightarrow \underline{d_{CARA} = \sqrt{2}a}$

030

Se considera que todas las cargas están fijas del infinito - $\left\{ U = \frac{K q_1 q_2}{r_{12}} \right\}$

CARGA N°1 -

$$U_1 = 0.5$$

Carga N°2 -

$$U_2 = \frac{K \cdot (+q) \cdot (-q)}{a} = -\frac{K q^2}{a} \quad (5)$$

Carga N°3

$$U_3 = \frac{K \cdot (+q) \cdot (-q)}{a} + \frac{K \cdot (-q) \cdot (-q)}{\sqrt{2} a}$$

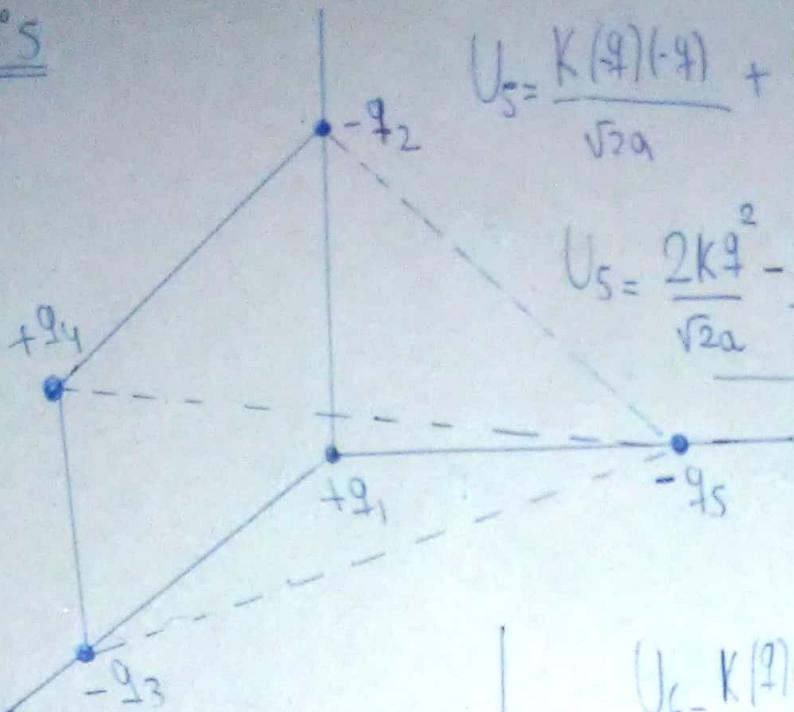
$$U_3 = \frac{K q^2}{a} \cdot \left(\frac{1}{\sqrt{2}} - \frac{1}{1} \right) \quad (5)$$

Carga N°4

$$U_4 = \frac{K \cdot (-q) \cdot (+q)}{a} + \frac{K \cdot (+q) \cdot (+q)}{\sqrt{2} a} + \frac{K \cdot (+q) \cdot (-q)}{a}$$

$$U_4 = \frac{-2K q^2}{a} + \frac{K q^2}{\sqrt{2} a} = \frac{K q^2}{a} \left(\frac{1}{\sqrt{2}} - 2 \right) \quad (5)$$

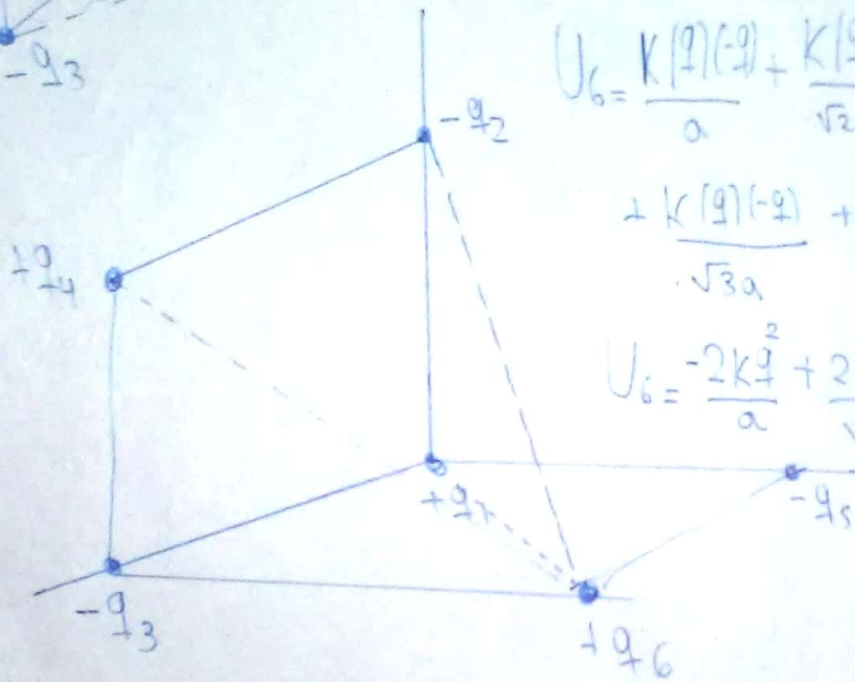
CARGA N° 5



$$U_5 = \frac{K(q)(-q)}{\sqrt{2}a} + \frac{K(-q)(q)}{\sqrt{3}a} + \frac{K(-q)(q)}{a} + \frac{K(-q)(-q)}{\sqrt{2}a}$$

$$U_5 = \frac{2Kq^2}{\sqrt{2}a} - \frac{Kq^2}{a} \left(\frac{1}{\sqrt{3}} + 1 \right)$$

Carga N° 6

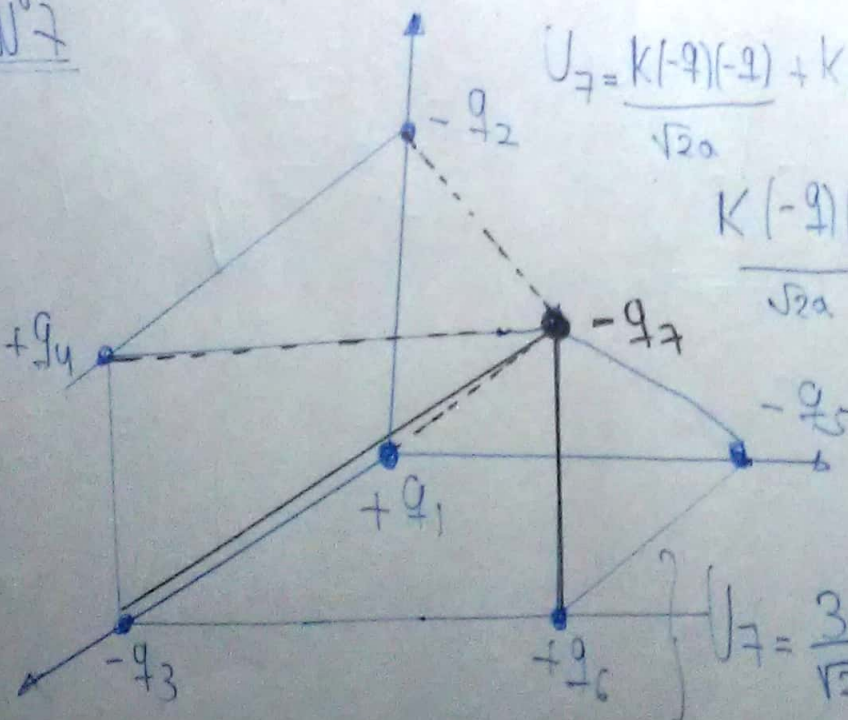


$$U_6 = \frac{K(q)(-q)}{a} + \frac{K(q)(q)}{\sqrt{2}a} + \frac{K(q)(-q)}{a}$$

$$+ \frac{K(q)(-q)}{\sqrt{3}a} + \frac{Kq(q)}{\sqrt{2}a}$$

$$U_6 = \frac{-2Kq^2}{a} + \frac{2Kq^2}{\sqrt{2}a} - \frac{Kq^2}{\sqrt{3}a}$$

Carga N° 7



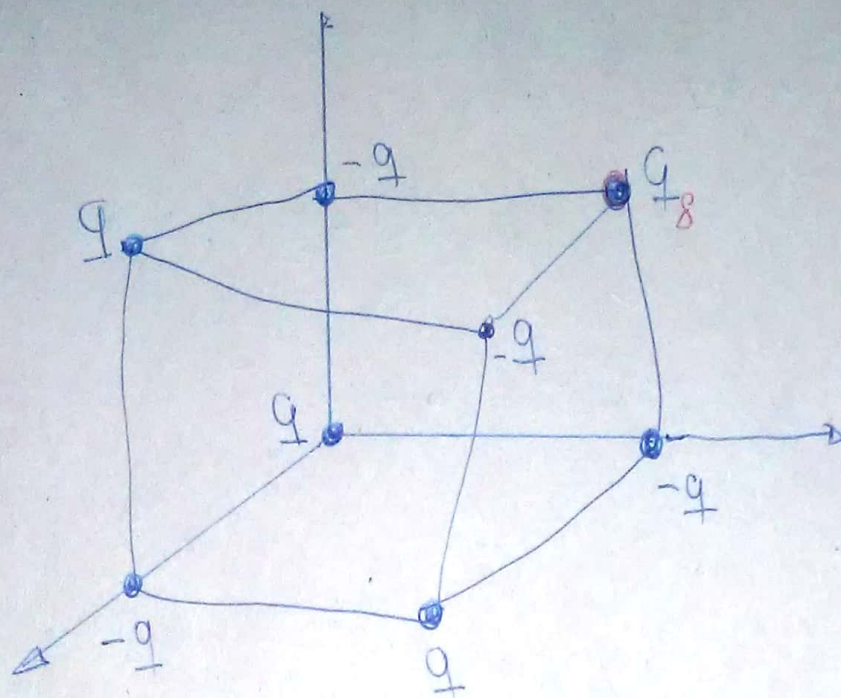
$$U_7 = \frac{K(-q)(-q)}{\sqrt{2}a} + \frac{K(-q)(q)}{a} + \frac{K(q)(-q)}{\sqrt{3}a} +$$

$$\frac{K(-q)(-q)}{\sqrt{2}a} + \frac{K(-q)(-q)}{\sqrt{2}a}$$

$$+ \frac{K(-q)(q)}{a}$$

$$U_7 = \frac{3Kq^2}{\sqrt{2}a} - \frac{Kq^2}{a} - \frac{Kq^2}{\sqrt{3}a}$$

Exercício 11.8



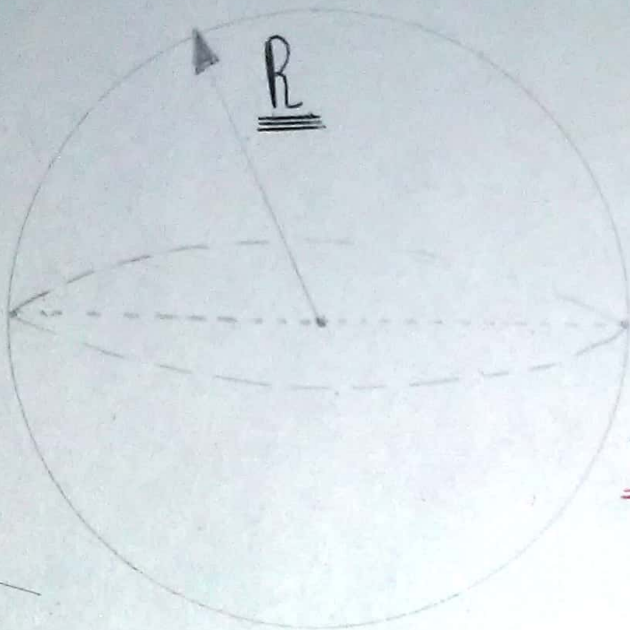
$$U_8 = \frac{k(-q)(q)}{a} + \frac{k(q)(q)}{\sqrt{2}a} + \frac{k(-q)(q)}{a} + \frac{k(q)(-q)}{a} + \frac{k(q)q}{\sqrt{2}a} \\ + \frac{k(q)(-q)}{\sqrt{3}a} + \frac{k(q)(q)}{\sqrt{2}a}$$

$$U_8 = -\frac{3kq^2}{a} + \frac{3kq^2}{\sqrt{2}a} - \frac{kq^2}{\sqrt{3}a}$$

$$U_T = \sum_{i=1}^8 (U_i) = -\frac{12kq^2}{a} \cdot \left(1 - \frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{9}\right) \cdot (5)$$

2) Determine la energía electrostática mecánica:

Habermos, que para una distribución volumétrica de la cantidad de carga eléctrica



$$U = \frac{1}{2} \cdot \left(\int_V V dQ \right) \dots (8')$$

donde ..

V: Potencial eléctrico de acuerdo a las coordenadas de dQ.

U: Energía potencial electrostática.

Habermos ..

$$\frac{dQ}{dV} = \rho_0 \Rightarrow dQ = \rho_0 \cdot dV, \text{ ahora como el volumen: } V = \frac{4\pi}{3} \cdot r^3 \Rightarrow dV = 4\pi r^2 dr \dots (2)$$

De (2) ..

$$dQ = \rho_0 \cdot (4\pi \cdot r^2 dr) \Rightarrow dQ = (4\pi \rho_0) r^2 dr \left\{ \begin{array}{l} \int dQ = (4\pi \rho_0) r^2 dr \\ \Rightarrow Q = \frac{4\pi \rho_0 r^3}{3} \dots (3) \end{array} \right.$$

(3) en (8') ..

$$U = \frac{1}{2} \cdot \left(\int_0^R V \cdot (4\pi \rho_0) \cdot r^2 dr \right) = \frac{4\pi \rho_0}{2} \cdot \left(\int_0^R V \cdot r^2 dr \right) = 2\pi \rho_0 \cdot \left(\int_0^R V \cdot r^2 dr \right) \dots (4)$$

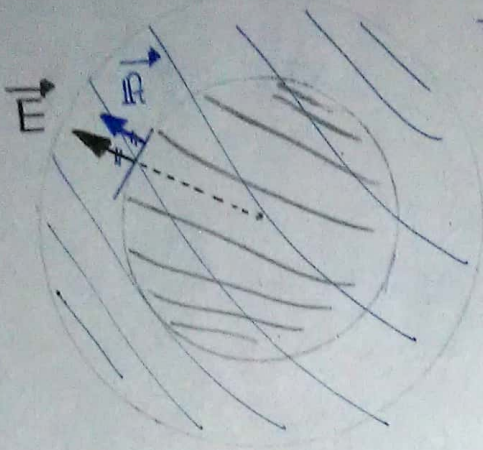
Habermos el potencial eléctrico ..

→ Primero habermos E ..

$$\text{Para } r < R: \oint_S \langle \vec{E}, d\vec{A} \rangle = \frac{Q_{\text{ENCERRADA}}}{\epsilon_0}$$

$$\Rightarrow E \cdot \left(\oint_S dA \right) = \frac{Q_{in}}{\epsilon_0}$$

$D_2(w)$



$$E \cdot 4\pi r^2 = \frac{\left(\frac{4\pi \rho_0 r^3}{3} \right)}{\epsilon_0}$$

$$E \cdot (4\pi r^2) = \frac{(4\pi \rho_0 r^3)}{3\epsilon_0}$$

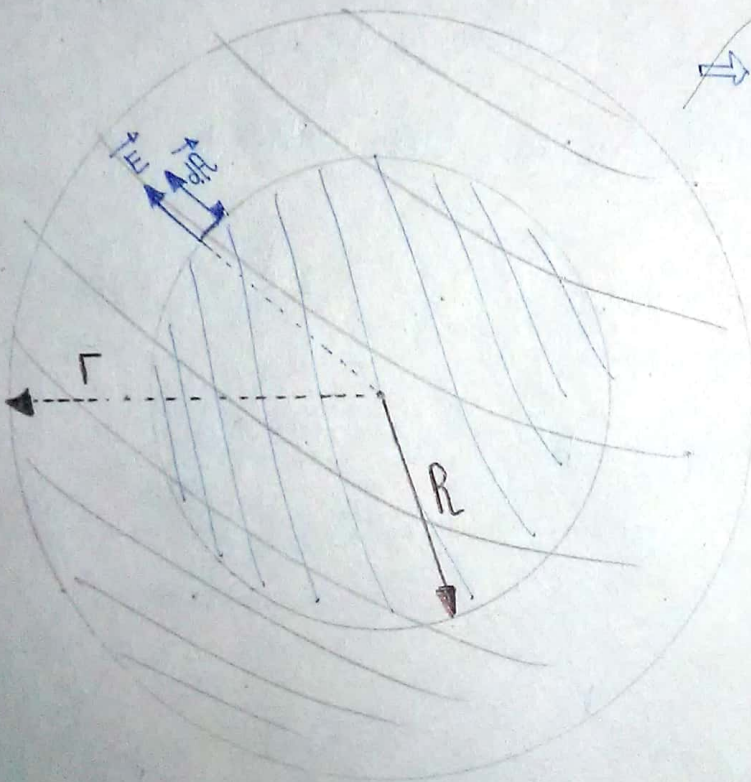
$$\therefore \underline{\underline{(\Gamma < 0; R)}} \dots \underline{\underline{E = \frac{\rho_0 \cdot \Gamma}{3\epsilon_0}}}$$

Cuando $\Gamma \geq R$

$$\Rightarrow E \cdot \left(\oint_S dA \right) = \frac{Q_{NETA (ESFERA)}}{\epsilon_0}$$

$$E \cdot (4\pi \Gamma^2) = \frac{(\rho_0 \cdot [\frac{4\pi R^3}{3}])}{\epsilon_0}$$

$$\underline{\underline{E = \frac{\rho_0}{3\epsilon_0} \cdot \left(\frac{R^3}{\Gamma^2} \right)}}$$



Sabemos : E : intensidad del campo eléctrico (MÓDULO)

$$E_{(\Gamma)} = \begin{cases} \frac{\rho_0 \cdot \Gamma}{3\epsilon_0}, & \Gamma \in (0; R) \\ \frac{\rho_0 \cdot R^3}{3\epsilon_0 \cdot \Gamma^2}, & \Gamma \in [R; +\infty) \end{cases}$$

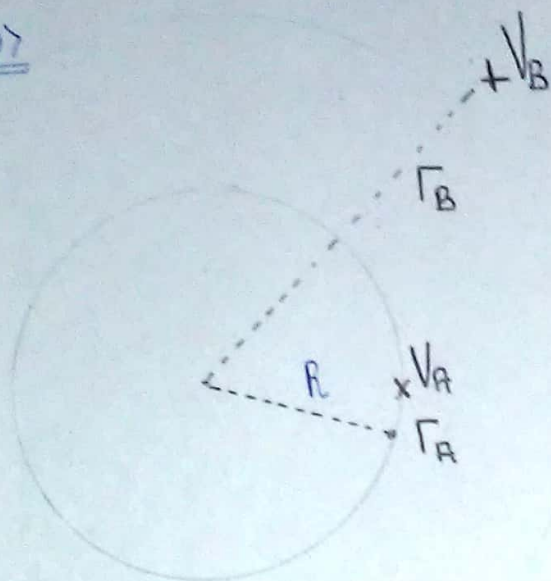
Ya teniendo esto, calculemos el potencial eléctrico.

Problem

$$V_B - V_A = - \int_A^B \langle \vec{E}; d\vec{l} \rangle$$

$$\rightarrow V_{B(\infty)} - V_A(r) = - \int_r^\infty \langle \vec{E}; d\vec{l} \rangle$$

$r \in [R; +\infty)$



$$V_{B(\infty)} - V_A(r) = - \int_r^\infty \left(\frac{\rho_0 R^3}{3\epsilon_0 r^2} \right) dr$$

$$V_{B(\infty)} - V_A(r) = + \int_r^\infty \left(\frac{\rho_0 R^3}{3\epsilon_0} \right) \left(\frac{-1}{r^2} \right) dr$$

$$-V_A(r) = \frac{\rho_0 R^3}{3\epsilon_0} \left(\left[\frac{1}{r} \right]_r^\infty \right)$$

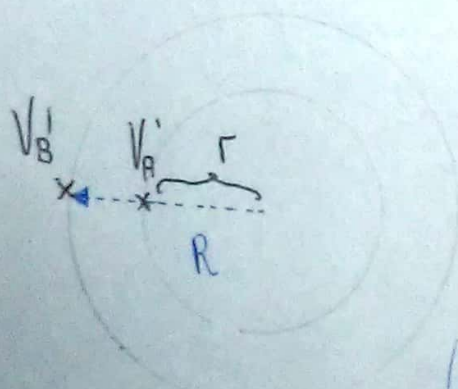
$r \in [R; +\infty)$

$$V_A(r) = \frac{\rho_0 R^3}{3\epsilon_0 r}$$

$$\left\{ -V_A(r) = \frac{\rho_0 R^3}{3\epsilon_0} \cdot \left(-\frac{1}{r} \right) \right.$$

$r \in (0; R)$

$$\rightarrow V_{B(R)} - V_A(r) = - \int_r^R \langle \vec{E}; d\vec{l} \rangle$$



$$\left(\frac{\rho_0 R^3}{3\epsilon_0 [R]} \right) - V_A(r) = - \int_r^R \left(\frac{\rho_0 r}{3\epsilon_0} \right) dr$$

$$\left(\frac{\rho_0 R^2}{3\epsilon_0} \right) - V_A(r) = - \frac{\rho_0}{3\epsilon_0} \left(\int_r^R r dr \right)$$

$$\left(\frac{\rho_0 R^2}{3\epsilon_0} \right) - V_A(r) = - \frac{\rho_0}{3\epsilon_0} \cdot \left(\frac{R^2}{2} - \frac{r^2}{2} \right)$$

$$\left. \begin{aligned} 2 \cdot \left(\frac{\rho_0 R^2}{3\epsilon_0} \right) - V_A(r) &= - \frac{\rho_0 R^2}{6\epsilon_0} + \frac{\rho_0 r^2}{6\epsilon_0} \end{aligned} \right\} V_A(r) = \frac{\rho_0 (3R^2 - r^2)}{6\epsilon_0}$$

En resumen

$$V(r) = \begin{cases} \frac{\rho_0 \cdot (3R^2 - r^2)}{6\epsilon_0}, & r \in [0; R] \\ \frac{\rho_0 \cdot R^3}{3\epsilon_0 \cdot r}, & r \in [R; +\infty) \end{cases}$$

De la situación (b)

$$U = 2\pi \cdot \rho_0 \cdot \left(\int_0^R \left(\frac{\rho_0}{6\epsilon_0} \cdot (3R^2 - r^2) \right) r \, dr \right)$$

$$U = \frac{\pi \cdot \rho_0^2}{3\epsilon_0} \cdot \left(\int_0^R (3R^2 r^2) \, dr - \int_0^R (r^4) \, dr \right)$$

$$U = \frac{\pi \cdot \rho_0^2}{3\epsilon_0} \cdot \left(3R^2 \cdot \left(\frac{r^3}{3} \Big|_0^R \right) - \left(\frac{r^5}{5} \Big|_0^R \right) \right)$$

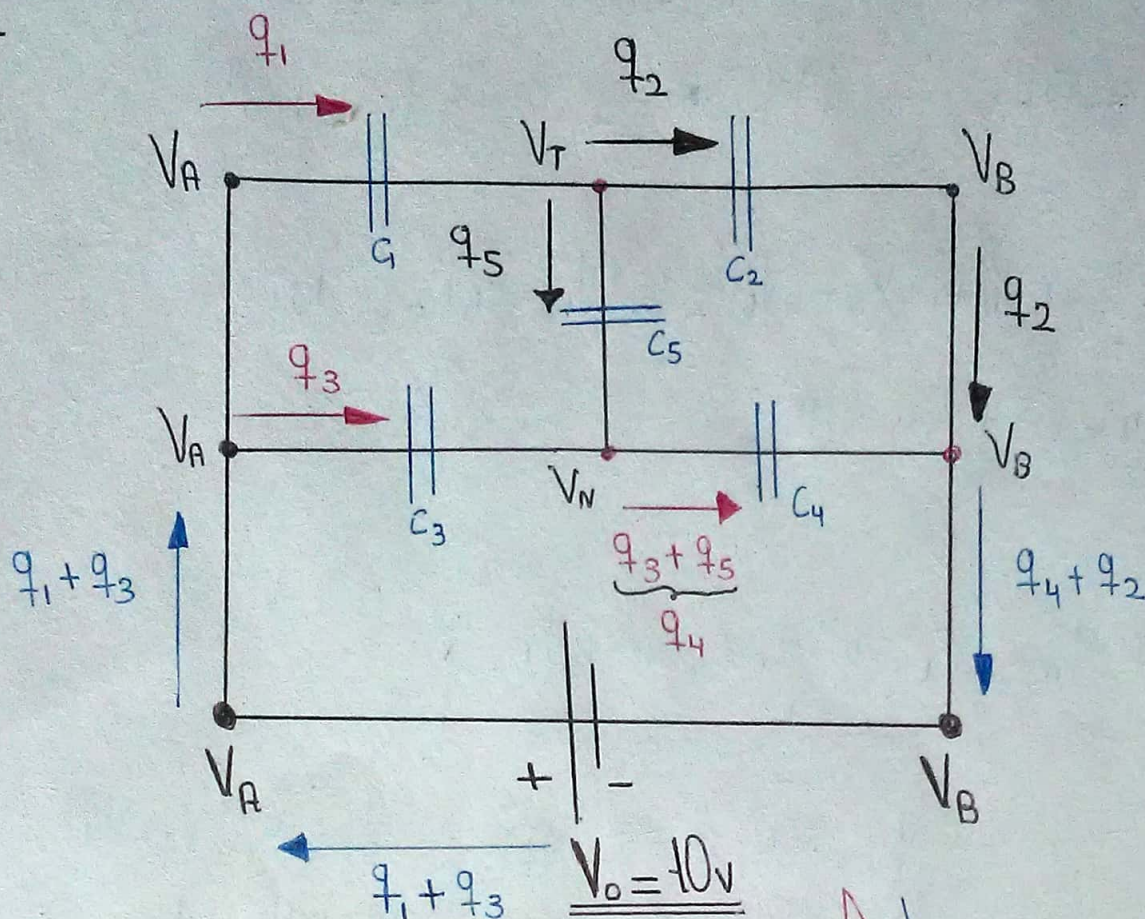
$$U = \frac{\pi \cdot \rho_0^2}{3\epsilon_0} \cdot \left(\cancel{3R^2} \cdot \frac{R^3}{\cancel{3}} - \frac{R^5}{5} \right) = \frac{\pi \cdot \rho_0^2}{3\epsilon_0} \cdot \left(\frac{5R^5}{5} - \frac{R^5}{5} \right)$$

$$U = \frac{4R^5 \pi \rho_0^2}{15 \epsilon_0} \quad (\text{J})$$

//

3) (Diagrama del circuito) ... $(C_1 = C_2 = 2 \mu F) \wedge (C_3 = C_4 = 6 \mu F) \wedge (C_5 = 5 \mu F)$

a)



Sabemos ... (Sentido Convencional de la Corriente Eléctrica)

$C = \frac{Q}{\Delta V}$: C ... Capacitancia
 Q ... Cantidad de Carga eléctrica
 ΔV ... Diferencia de Potencial Eléctrico

Sabemos ...

Por la ley de la conservación de la cantidad de carga eléctrica ...

$$Q_i = Q_f \begin{cases} Q_i: \text{Cantidad de carga meta al inicio.} \\ Q_f: \text{Cantidad de carga meta al final.} \end{cases}$$

En base a ello, establecemos ...

$$C_1 = \frac{q_1}{\Delta V_1} = \frac{q_1}{V_A - V_T} = 2 \mu F \Rightarrow \underline{q_1 = 2V_A - 2V_T} \dots (1)$$

$$C_2 = \frac{q_2}{\Delta V_2} = \frac{q_2}{V_T - V_B} = 2 \mu F \Rightarrow \underline{q_2 = 2V_T - 2V_B} \dots (2)$$

$$C_3 = \frac{q_3}{\Delta V_3} = \frac{q_3}{V_A - V_N} = 6 \mu F \Rightarrow \underline{q_3 = 6V_A - 6V_N} \dots (3)$$

$$C_5 = \frac{q_5}{\Delta V_5} = \frac{q_5}{V_T - V_N} = 5 \mu F \Rightarrow \underline{q_5 = 5V_T - 5V_N} \dots (5)$$

$$C_4 = \frac{q_4}{\Delta V_4} = \frac{q_4}{V_N - V_B} = 6 \mu F \Rightarrow \underline{q_4 = 6V_N - 6V_B} \dots (4)$$

OJO ...

$$V_0 = 10V \Rightarrow \underline{V_A - V_B = 10V}$$

(1)+(2) ..

$$q_1 + q_2 = 2V_A - \cancel{2V_T} + (\cancel{2V_T} - 2V_B) = 2.(V_A - V_B) = 2.(10)$$

$$\Rightarrow \underline{q_1 + q_2 = 20 \mu C} \dots (d) \rightsquigarrow \underline{q_2 = 20 - q_1} \dots (d')$$

(3)+(4) ..

$$q_3 + q_4 = 6V_A - \cancel{6V_N} + (\cancel{6V_N} - 6V_B) = 6.(V_A - V_B) = 6.(10)$$

$$\Rightarrow \underline{q_3 + q_4 = 60 \mu C} \dots (p)$$

(1) \wedge (3) \wedge (5) ..

$$3.(q_1) = 3.(2V_A - 2V_T) \Rightarrow 3q_1 = 6V_A - 6V_T$$

$$\underline{q_3 = 6V_A - 6V_N} \quad \uparrow (-)$$

$$\downarrow \quad q_3 - 3q_1 = 6.(V_T - V_N) \quad \wedge \quad q_5 = 5.(V_T - V_N)$$

$$q_3 - 3q_1 = 6.\left(\frac{q_5}{5}\right)$$

$$\underline{V_T - V_N = \frac{q_5}{5}}$$

$$\underline{5q_3 - 15q_1 = 6q_5} \dots (q_1)$$

De la conservación de la carga meta ..

De (d)+(p) ..

$$\sum Q_{\text{SALIENTE}} = \sum Q_{\text{ENTRANTE}} \dots$$

$$\checkmark \underline{(q_1 + q_3) = (q_2 + q_4) = L}$$

$$\begin{cases} (q_1 + q_2) + (q_3 + q_4) = 80 \mu C \\ (q_1 + q_3) + (q_2 + q_4) = 80 \mu C \\ 2L = 80 \mu C \Rightarrow \underline{L = 40 \mu C} \end{cases}$$

$$\therefore \underline{q_1 + q_3 = 40 \mu C} \dots (q_2)$$

$$\underline{q_2 + q_4 = 40 \mu C} \dots (q')$$

$$\checkmark \underline{q_1 = q_2 + q_5}$$

$$q_1 - q_2 = q_5$$

De (d')

$$\dots q_1 - (20 - q_1) = q_5$$

$$2q_1 - 20 = q_5 \dots (q_3)$$

D_e (0₁)₁ (0₃) ..

$$5q_3 - 15q_1 = 6 \cdot (12q_1 - 20) = 12q_1 - 120$$

$$15q_1 + 12q_1 - 5q_3 = 120 \quad \left\{ \quad \underline{\underline{27q_1 - 5q_3 = 120}} \right.$$

D_e (0₂) .. $q_1 + q_3 = 40 \Rightarrow \underline{\underline{5q_1 + 5q_3 = 200}}$ (+)

$$(27q_1 - 5q_3) + (5q_1 + 5q_3) = 320$$

$$32q_1 = 320 \Rightarrow \underline{\underline{q_1 = 10 \mu C}} \quad \wedge \quad q_3 = 40 - q_1 = 40 - (10)$$

$$\Rightarrow \underline{\underline{q_3 = 30 \mu C}}$$

D_e (0₃)

$$2q_1 - 20 = q_5 \Rightarrow 2 \cdot (10) - 20 = 0 = q_5 \Rightarrow \underline{\underline{q_5 = 0 \mu C}}$$

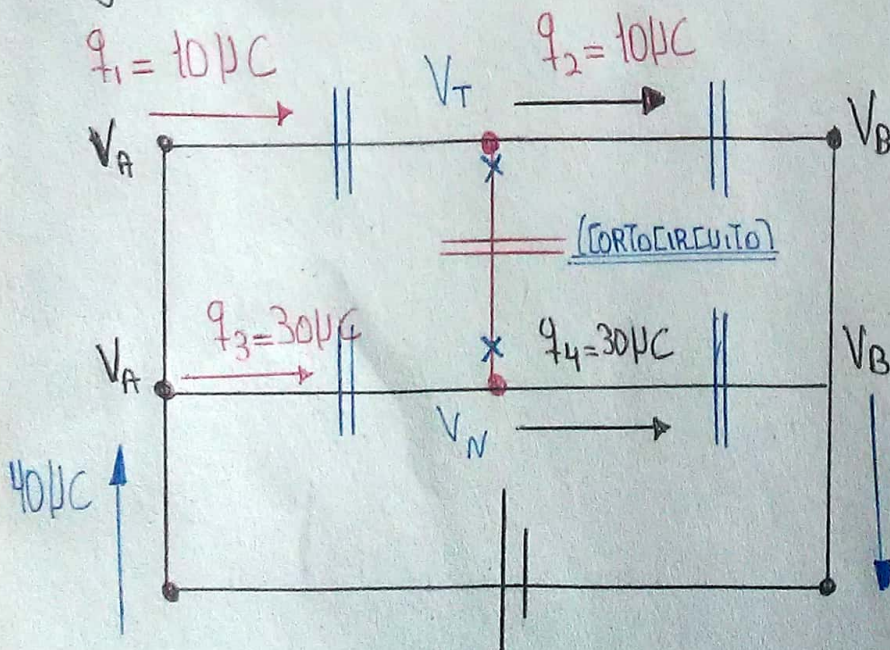
D_e (α) ..

$$q_1 + q_2 = 20 \mu C \Rightarrow (10 \mu C) + q_2 = 20 \mu C \Rightarrow \underline{\underline{q_2 = 10 \mu C}}$$

D_e (γ') ..

$$q_2 + q_4 = 40 \mu C \Rightarrow (10 \mu C) + q_4 = 40 \mu C \quad \left\{ \quad \underline{\underline{q_4 = 30 \mu C}} \right.$$

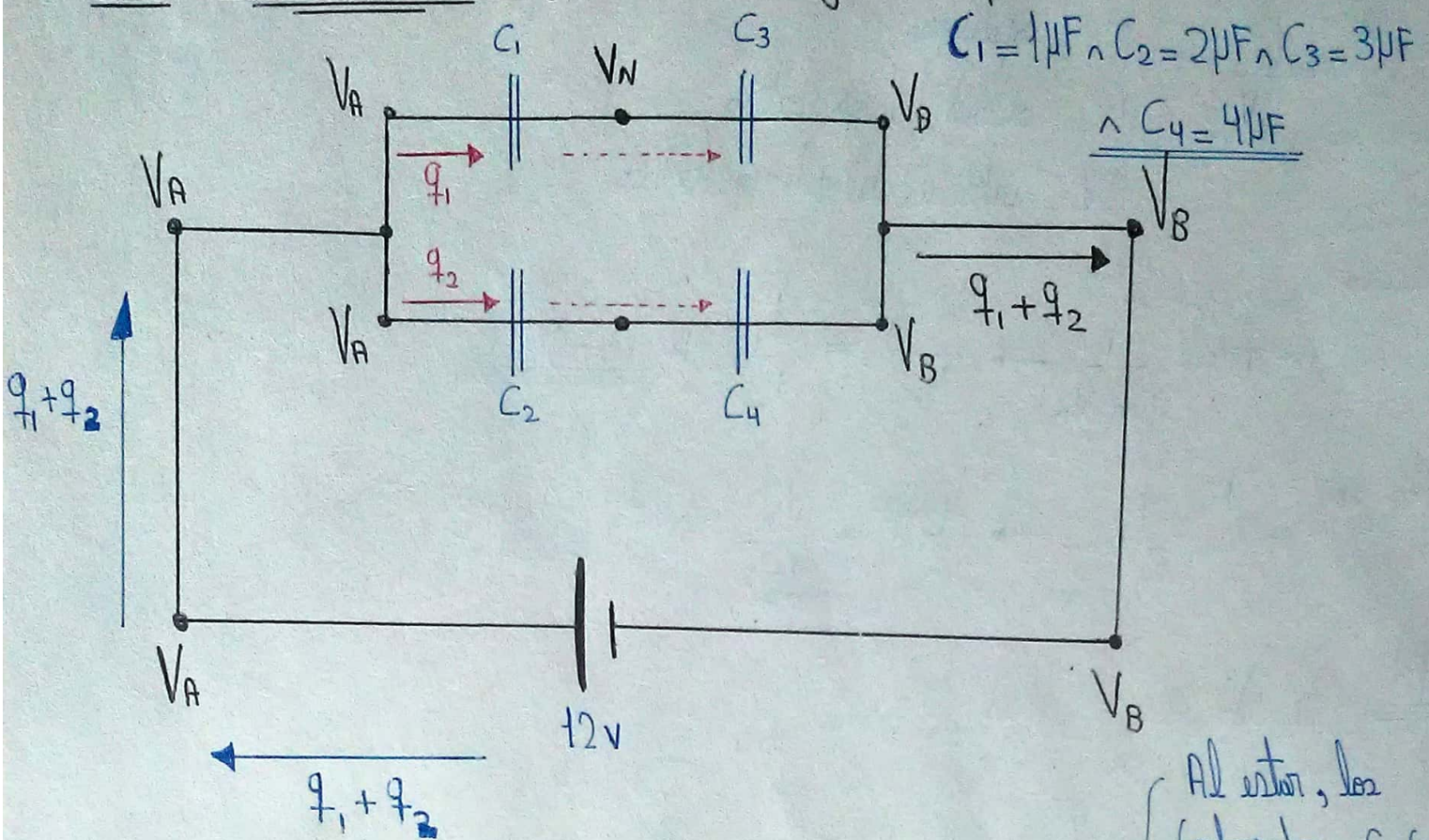
Diagrama Resultante ..



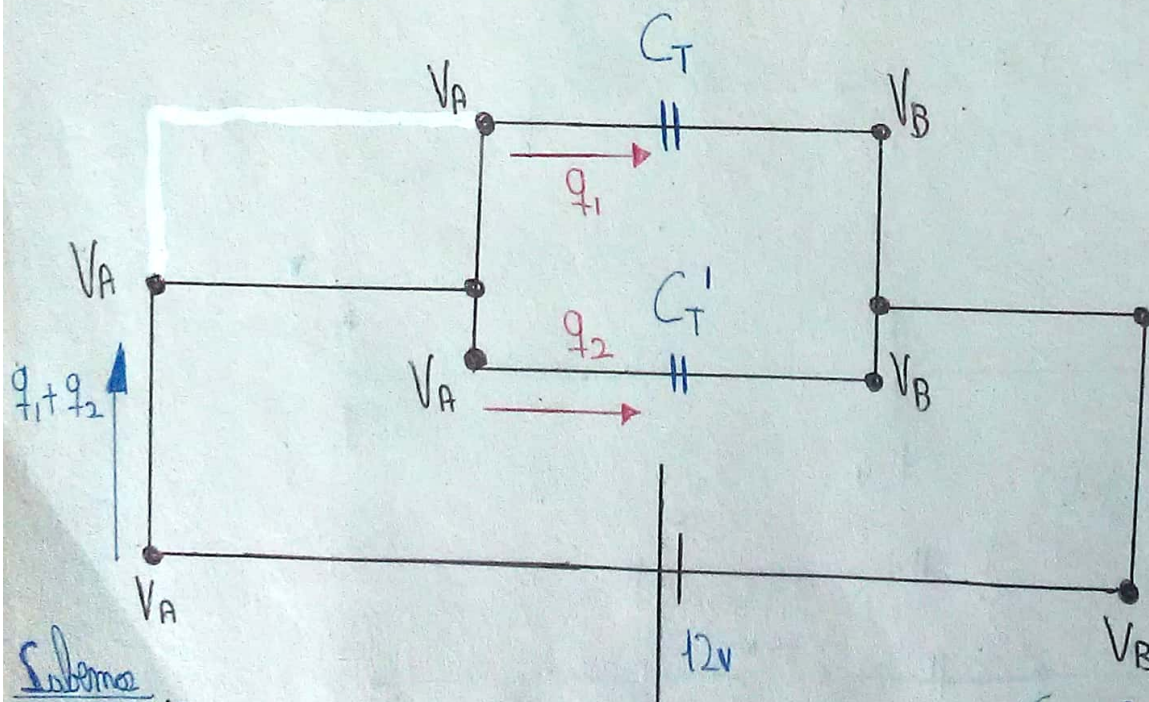
b) Debemos ..

Que el cortocircuito dado en el condensador C_5 es dado cuando en ambos extremos se tiene el mismo potencial eléctrico por lo que la diferencia del potencial $\underline{\underline{\Delta V_5 = 0V}}$

4) a) Cuando S_1 se cierra, se tiene el siguiente esquema. -



Se tiene, el siguiente sistema equivalente



Al estar, los condensadores C_1, C_3 en serie, la cantidad de carga que circula por ellos es q_1 . Por lo, C_2 y C_4 llevan q_2 .

Solamos

La Capacitancia equivalente es $C_T = \frac{C_1 \cdot C_2}{C_1 + C_2}$

$C_T = \frac{C_1 \cdot C_3}{C_1 + C_3} = \frac{(1\mu F)(3\mu F)}{(1\mu F + 3\mu F)} = \frac{3}{4} \mu F$

$C_T' = \frac{C_2 \cdot C_4}{C_2 + C_4} = \frac{(2\mu F)(4\mu F)}{(2\mu F + 4\mu F)} = \frac{4}{3} \mu F$

Problemas . .

$$\underline{\underline{C = \frac{Q}{\Delta V}}}$$

C . . Capacitancia

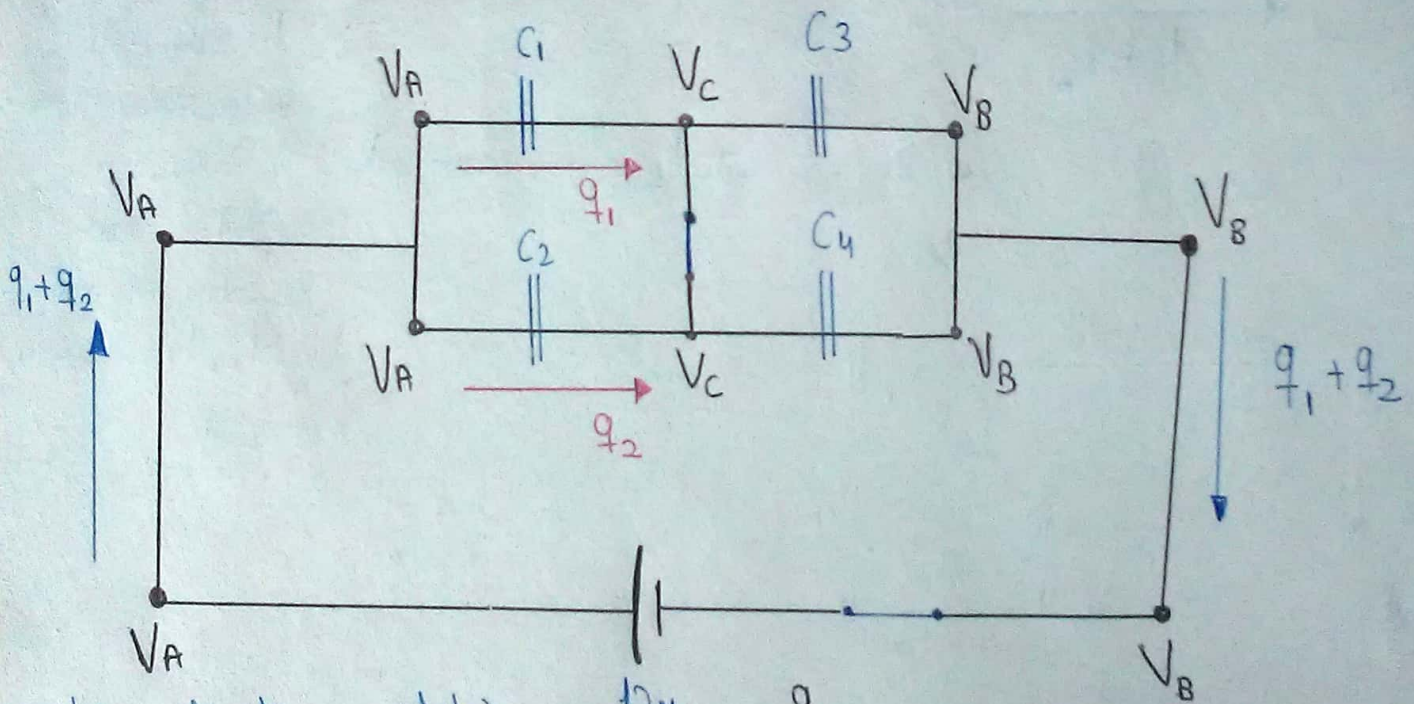
Q . . Cantidad de carga eléctrica

ΔV . . Diferencia del potencial eléctrico

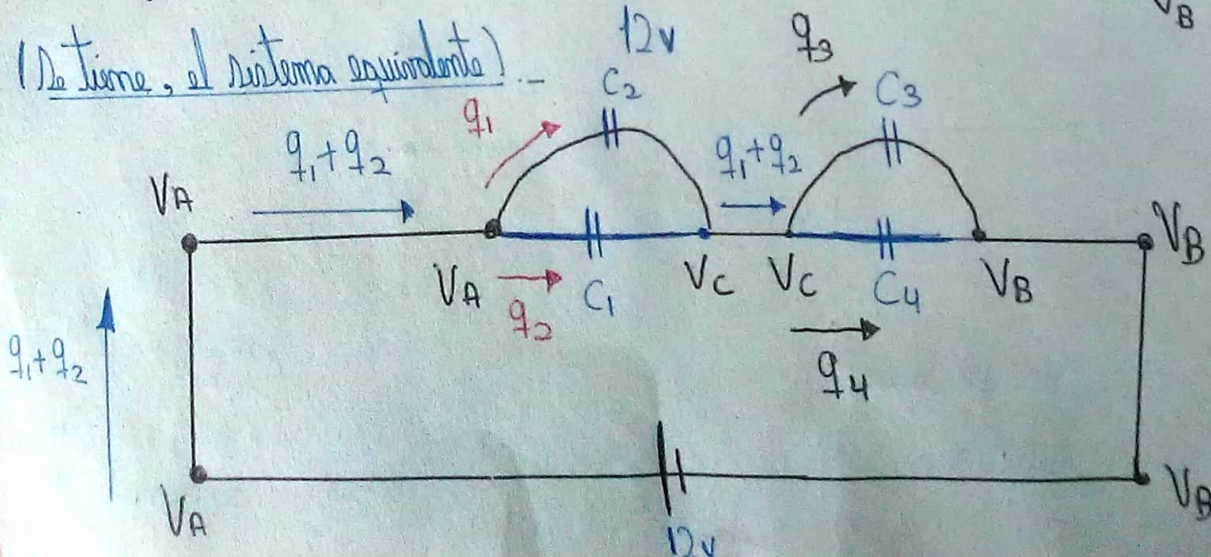
$$\rightarrow C_T = \frac{q_1}{(V_A - V_B)} \rightarrow \frac{3}{4} \mu F = \frac{q_1}{12} \Rightarrow \underline{\underline{q_1 = 9 \mu C}}$$

$$\rightarrow C_T' = \frac{q_2}{(V_A - V_B)} \rightarrow \frac{4}{3} \mu F = \frac{q_2}{12} \Rightarrow \underline{\underline{q_2 = 16 \mu C}}$$

b) Si además, se cierra S_2 . .



(De nuevo, el sistema equivalente) . .



$$\rightarrow C_1 = \frac{q_1}{V_A - V_C} = 1\mu F \Rightarrow \underline{q_1 = V_A - V_C} \dots (d_1)$$

$$\rightarrow C_2 = \frac{q_2}{V_A - V_C} = 2\mu F \Rightarrow \underline{q_2 = 2V_A - 2V_C} \dots (d_2)$$

$$\rightarrow C_3 = \frac{q_3}{V_C - V_B} = 3\mu F \Rightarrow \underline{q_3 = 3V_C - 3V_B} \dots (d_3)$$

$$\rightarrow C_4 = \frac{q_4}{V_C - V_B} = 4\mu F \Rightarrow \underline{q_4 = 4V_C - 4V_B} \dots (d_4)$$

Conservación de la Carga
meta eléctrica

$$\underline{q_1 + q_2 = q_3 + q_4} \dots (2)$$

Solemos ..

$$\underline{V_A - V_B = 12V} \dots (3)$$

De (d1, d2, d3, d4) ..

$$(V_A - V_C) + (2V_A - 2V_C) = (3V_C - 3V_B) + (4V_C - 4V_B)$$

$$3V_A - 3V_C = 7V_C - 7V_B \leadsto \underline{10V_C = 3V_A + 7V_B} \dots (8')$$

De (d1) ..

$$q_1 = V_A - V_C \Rightarrow 10q_1 = 10V_A - 10V_C \xrightarrow{(8')} 10q_1 = 10V_A - 3V_A - 7V_B$$

$$10q_1 = 7V_A - 7V_B = 7 \cdot (V_A - V_B)$$

$$q_1 = \frac{7}{10} \cdot (V_A - V_B)$$

De (3) .. $\left(q_1 = \frac{7}{10} \cdot (12) = \frac{42}{5} \mu C = \underline{8,4 \mu C} \right)$

De (d2) ..

$$q_2 = 2V_A - 2V_C = 2V_A - 2 \cdot (V_C) \Rightarrow 5q_2 = 10V_A - 10V_C = 10V_A - (3V_A + 7V_B)$$

$$5q_2 = 7 \cdot (V_A - V_B) \leadsto \left(q_2 = \frac{7}{5} \cdot (V_A - V_B) = \frac{7}{5} \cdot (12) = \underline{16,8 \mu C} \right)$$

D₂(23) ..

$$q_3 = 3V_C - 3V_B \Rightarrow \frac{10q_3}{3} = 10V_C - 10V_B = (3V_A + 7V_B - 10V_B)$$

$$\frac{10q_3}{3} = 3V_A - 3V_B = 3 \cdot (V_A - V_B) \Rightarrow q_3 = \frac{9}{10} \cdot (V_A - V_B) = \frac{9 \cdot 12}{10} = \underline{\underline{10,8 \mu C}}$$

D₂(24) ..

$$q_4 = 4V_C - 4V_B \Rightarrow \frac{10q_4}{4} = 10V_C - 10V_B = (3V_A + 7V_B) - 10V_B = 3(V_A - V_B)$$

$$q_4 = \frac{6}{5} \cdot (V_A - V_B) = \frac{6 \cdot 12}{5} = \underline{\underline{14,4 \mu C}}$$